

PRAKTIKUM SISTEM OPERASI

MODUL

(Syscall_Xinu)



Oleh:

Farhan Kurniawan (2311104073)

PROGRAM STUDI REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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Jurnal/TP

Soal:

1. [50 Poin] Buat syscall baru seperti yang ditunjukkan pada modul syscall poin 9.5! (sertakan Screenshot kode dan hasil run)

2. [25 Poin] Perbaiki syscall chprio (xinu/system/chprio.c) dengan memperhatikan validasi input
 - Pastikan id adalah angka dari 0 – NPROC (ukuran maks banyaknya proses)
 - Pastikan prioritas adalah bilangan yang positif

Compile dan jalankan Xinu dengan syscall yang telah diperbaiki

- make clean
- make

3. Lakukan hal-hal berikut ini

- Edit xsh_uptime.c
- Tambahkan kode berikut

```
/* Check for valid number of arguments */
if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

// tambahan code
chprio(5,33) // menanggil syscall chprio; mengubah prioritas menjadi 33 pada id=5
```

- Compile source code tersebut dengan perintah
 - make clean
 - make
- Jalankan perintah ps
 - xsh \$ ps
 - perhatikan prioritas proses dengan id = 5
- Jalankan uptime
 - xsh \$ uptime
 - Perhatikan hasil perintah tersebut
- Jalankan ps
 - xsh \$ ps
 - perhatikan prioritas proses dengan id = 5 seharusnya sudah berubah

[25 Poin] Testing chprio syscall yang telah diubah

- Testing prioritas tidak boleh < 0: Ubah “chprio(5,33)” menjadi “chprio(5,-3)” pada xsh_uptime.c
- Testing id adalah valid: Ubah “chprio(5,33)” menjadi “chprio(3000,3)”
- Hasil dua testing di atas adalah prioritas tidak berubah karena salah argument (dibuktikan dengan menggunakan perintah ps)

Jawaban

1. Pada praktikum ini dibuat syscall baru bernama `chname()`. Syscall tersebut menampilkan pesan **"New Syscall is easy"** ketika dipanggil. Setelah menambahkan deklarasi pada `prototypes.h` dan memanggil syscall pada `xsh_uptime.c`, syscall berhasil dijalankan saat perintah `uptime` dieksekusi.

Input:

system/chname.c



```
Activities gedit Tue 16 Jun, 12:24
chname.c ~/xnu/system
Save

/*
chname.c
Farhan Kurniawan - 2311104073
*/

#include <xinu.h>

syscall chname(void)
{
    intmask mask;

    mask = disable();

    kprintf("New syscall is easy\n");

    restore(mask);
    return OK;
}
```

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include/prototypes.h



```
/* in file userret.c */
extern void userret(void);

/* in file wait.c */
extern syscall wait(sid32);

/* in file wakeup.c */
extern void wakeup(void);

/* in file write.c */
extern syscall write(did32, char *, uint32);

/* in file xdone.c */
extern void xdone(void);

/* in file yield.c */
extern syscall yield(void);
extern syscall chname(void);

/* NETWORK BYTE ORDER CONVERSION NOT NEEDED ON A BIG-ENDIAN COMPUTER */
#define htons(x) ( ((x)>>8) | ((x)<<8) )
#define htonl(x) ( (((x)>>24) & 0x000000ff) | (((x)>> 8) & 0x0000ff00) | \
  (((x)<< 8) & 0x00ff0000) | (((x)<<24) & 0xff000000) )
#define ntohs(x) ( ((x)>>8) | ((x)<<8) )
#define ntohl(x) ( (((x)>>24) & 0x000000ff) | (((x)>> 8) & 0x0000ff00) | \
  (((x)<< 8) & 0x00ff0000) | (((x)<<24) & 0xff000000) )
```

shell/xsh_uptime.c

```
Activities gedit Tue 16 Jun, 12:27
xsh_uptime.c
~/xinu/shell
Save
Open
xsh_uptime.c - xsh_uptime */
/* Nama: Farhan Kurniawan */
/* NIM: 2311104073 */
#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_uptime - shell to print the time the system has been up
 *-----
 */
shellcmd xsh_uptime(int nargs, char *args[])
{
    chname();
    uint32 days, hrs, mins, secs; /* Days, hours, minutes, and */
                                /* seconds since system boot */
    uint32 secperday = 86400; /* Seconds in a day */
    uint32 secperhr = 3600; /* Seconds in an hour */
    uint32 secpermin = 60; /* Seconds in a minute */

    /* For argument '--help', emit help about the 'uptime' command */

    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n\n", args[0]);
        printf("Description:\n");
        printf("\tDisplays time since the system booted\n");
        printf("Options:\n");
        printf("\t--help\t display this help and exit\n");
    }
}
```

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Output:

```
Activities Terminal Tue 16 Jun, 12:22
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help
[0x00100000 to 0x005FBFFF]
115411 bytes of Xinu code.
[0x00100000 to 0x0011C2D2]
148392 bytes of data.
[0x0011FF20 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)

-----FarhanKurniawan-----

XINU

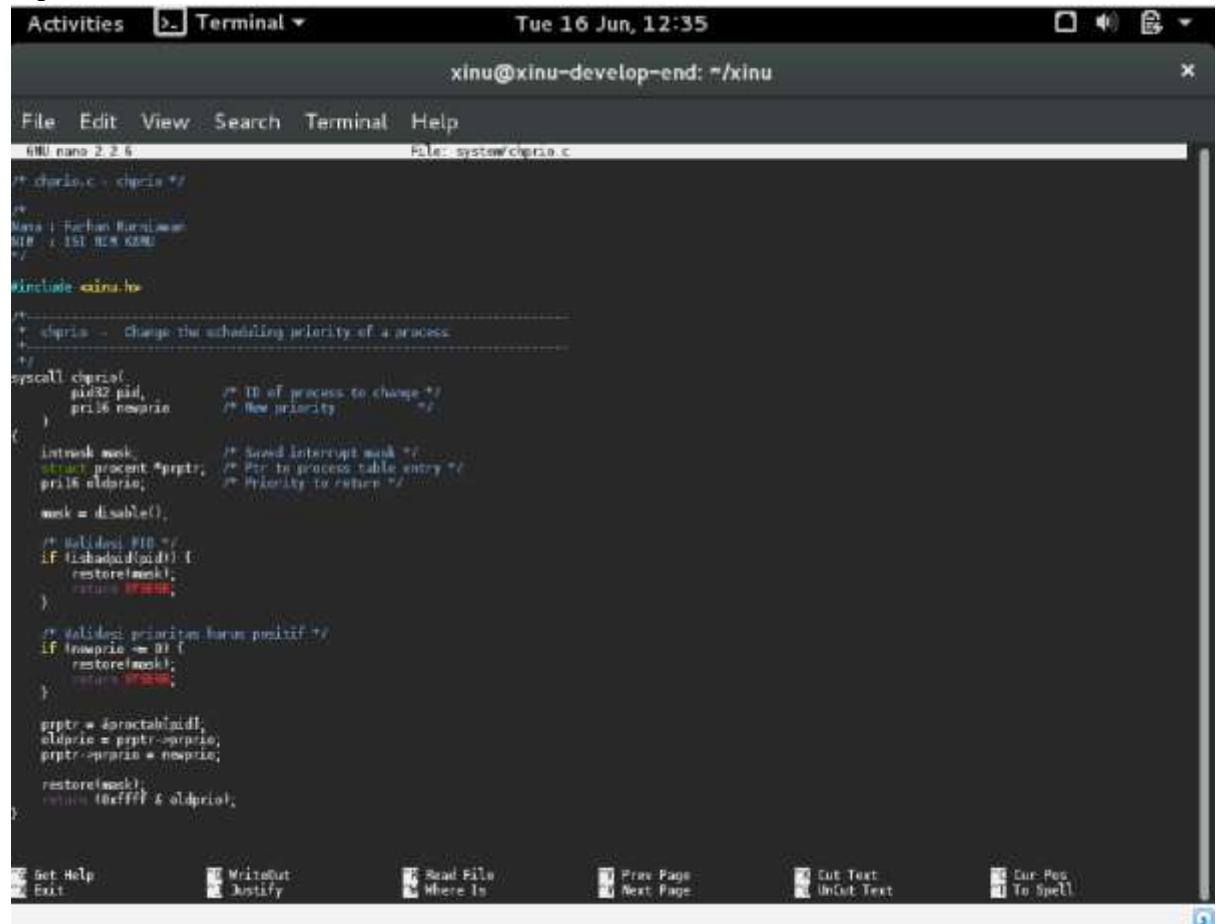
-----2311104073-----

Welcome to Xinu!

xsh $ uptime
New syscall is easy
Uptime = 64 seconds
New syscall is easy
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:2 | ttyS0
```

2. Pada percobaan ini dilakukan perbaikan terhadap syscall `chprio()` dengan menambahkan validasi input. Validasi dilakukan untuk memastikan PID berada dalam rentang yang valid dan prioritas bernilai positif. Setelah dilakukan kompilasi ulang, syscall dapat bekerja lebih aman karena menolak input yang tidak sesuai. Dengan demikian, perubahan prioritas hanya dapat dilakukan jika parameter yang diberikan benar.

Input:



```
Activities Terminal Tue 16 Jun, 12:35
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
GNU nano 2.2.6 File: system/chprio.c

/* chprio.c - chprio */
/*
 * Name : Farhan Ruzulwan
 * NIM : 151 019 6282
 */

#include <stdio.h>

/* chprio - change the scheduling priority of a process */
/*
 */

syscall chprio(
    pid32 pid, /* ID of process to change */
    pri16 newprio /* New priority */
)
{
    intmask mask; /* Saved interrupt mask */
    struct procnt *pptr; /* Ptr to process table entry */
    pri16 oldprio; /* Priority to return */

    mask = disable();

    /* Validasi PID */
    if (!isvalid(pid)) {
        restore(mask);
        return -EINVAL;
    }

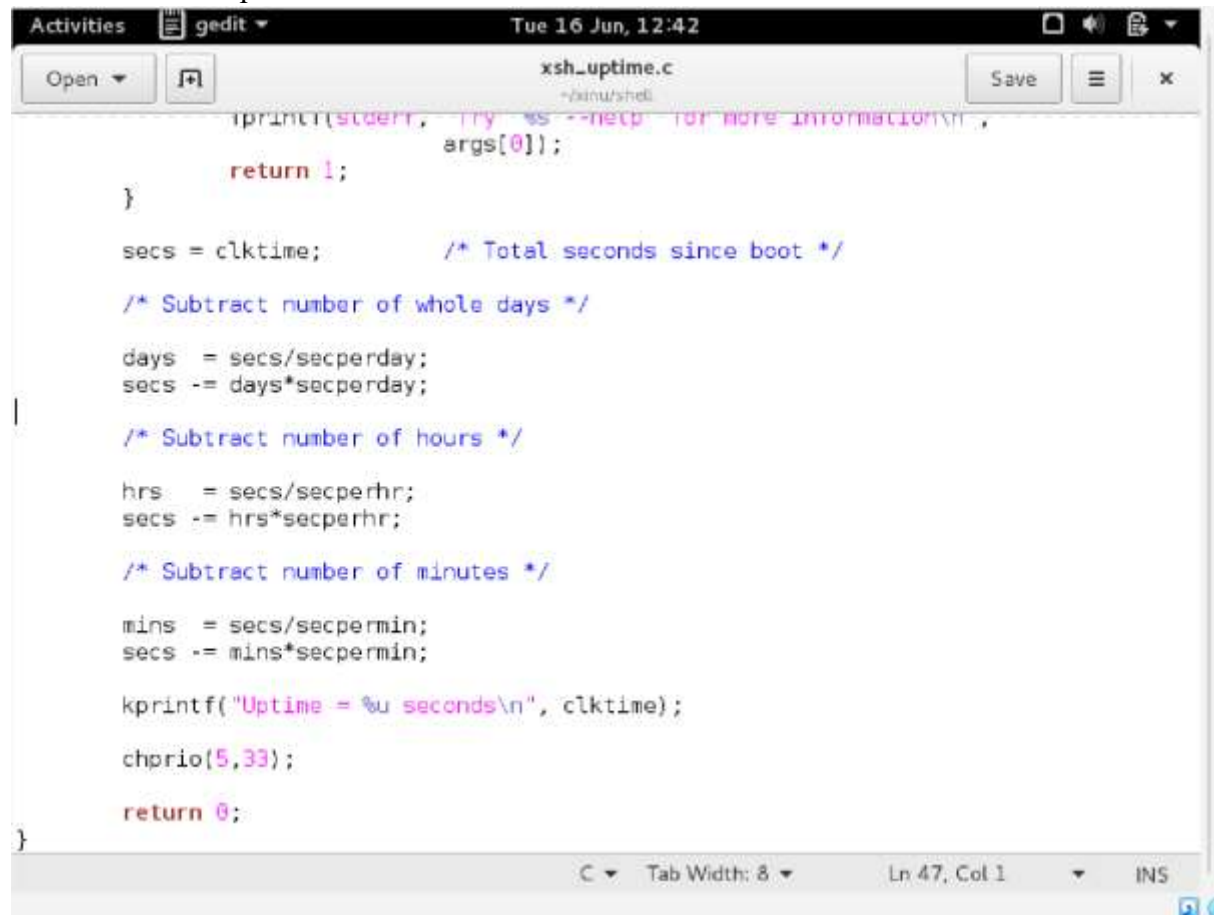
    /* Validasi prioritas harus positif */
    if (newprio <= 0) {
        restore(mask);
        return -EINVAL;
    }

    pptr = dprocab(pid);
    oldprio = pptr->oprio;
    pptr->oprio = newprio;

    restore(mask);
    return (0xffff & oldprio);
}
```

3. Pada percobaan ini syscall `chprio()` digunakan untuk mengubah prioritas proses dengan PID tertentu. Hasil perintah `ps` sebelum dan sesudah menjalankan uptime menunjukkan bahwa prioritas proses berhasil berubah sesuai nilai yang diberikan. Selanjutnya dilakukan pengujian menggunakan prioritas negatif dan PID yang tidak valid. Hasil pengujian menunjukkan bahwa prioritas proses tidak berubah karena input ditolak oleh validasi yang telah ditambahkan.

Prioritas setelah uptime:



```
iprintf(stderr, "Try 'ps -help' for more information\n",
        args[0]);
    return 1;
}

secs = clktime;      /* Total seconds since boot */

/* Subtract number of whole days */

days = secs/secperday;
secs -= days*secperday;

/* Subtract number of hours */

hrs = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */

mins = secs/secpermin;
secs -= mins*secpermin;

kprintf("Uptime = %u seconds\n", clktime);

chprio(5, 33);

return 0;
}
```



```
Activities Terminal Tue 16 Jun, 12:48
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

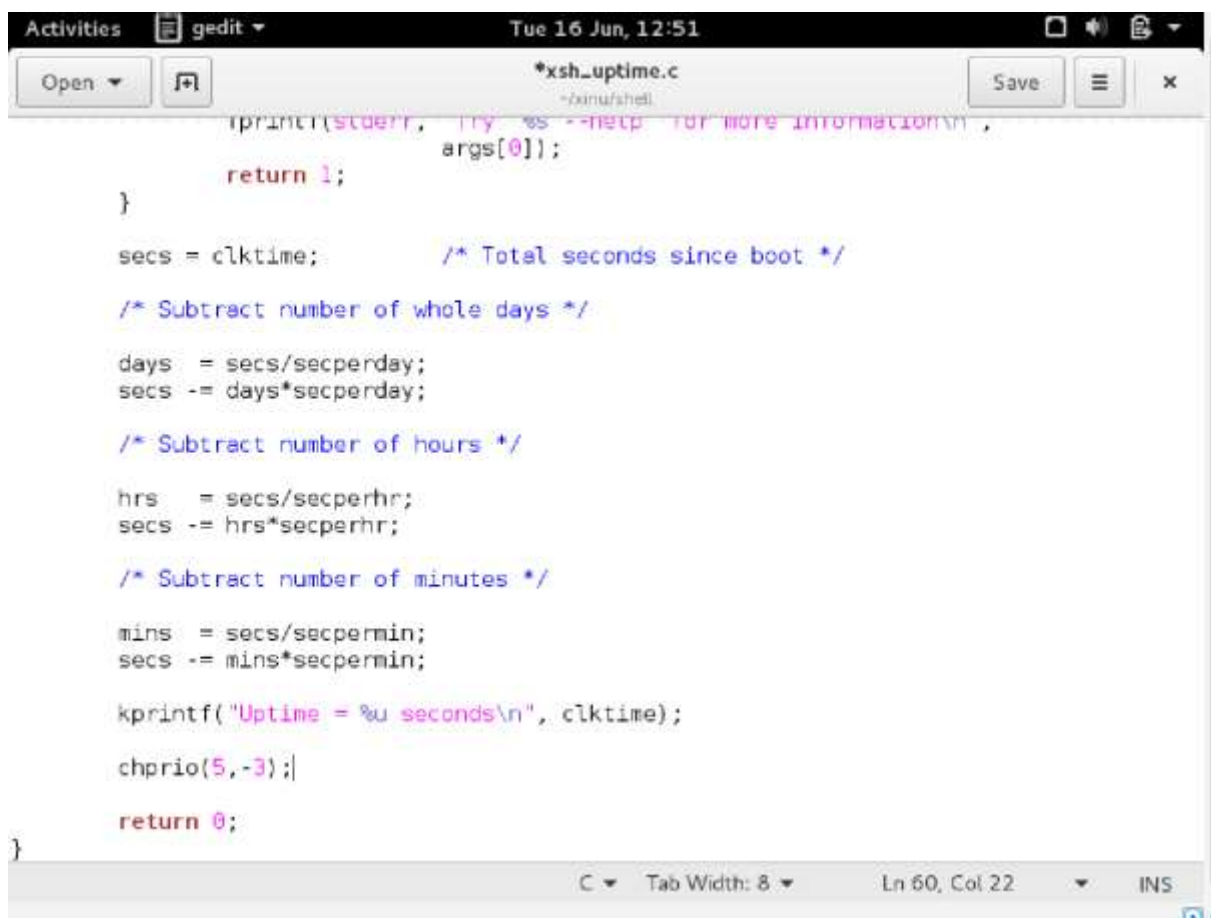
XINU
-----2311104073-----

Welcome to Xinu!

xsh $ ps
Pid Name State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 pnull ready 0 0 0x005FDFFC 0x00FFFEF0 8192
1 3/n 1 ipout wait 500 0 0x005FBFFC 0x005FBF30 8192
0 0/n 2 netin wait 500 0 0x005F9FFC 0x005F9ED0 8192
0 0/n 4 Main process sleep 20 3 0x005E7FFC 0x005E7F44 65536
0 0/n 5 shell recv 50 4 0x005D7FFC 0x005D79AC 8192
0 0/n 6 ps curr 20 5 0x005F7FFC 0x005F7FC4 8192
0 0/nxsh $ uptime
New syscall is easy
Uptime = 68 seconds
xsh $ uptime
New syscall is easy
Uptime = 157 seconds
xsh $

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:3 | ttyS0
```

Prioritas Negatif:



```
Activities gedit Tue 16 Jun, 12:51
*xsh_uptime.c
~binu/shell
Open Save
fprintf(stderr, "Try 'xsh -help' for more information\n",
        args[0]);
    return 1;
}

secs = clktime;      /* Total seconds since boot */

/* Subtract number of whole days */

days = secs/secperday;
secs -= days*secperday;

/* Subtract number of hours */

hrs = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */

mins = secs/secpermin;
secs -= mins*secpermin;

kprintf("Uptime = %u seconds\n", clktime);

chprio(5, -3);

return 0;
}
```

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```
Activities Terminal Tue 16 Jun, 12:54
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192
1             3/n  1 ipout wait  500  0 0x005FBFFC 0x005FBF30 8192
0             0/n  2 netin wait  500  0 0x005F9FFC 0x005F9ED0 8192
0             0/n  4 Main process sleep 20  3 0x005E7FFC 0x005E7F44 65536
0             0/n  5 shell  recv  50  4 0x005D7FFC 0x005D79AC 8192
0             0/n  6 ps    curr  20  5 0x005F7FFC 0x005F7FC4 8192
0             0/nxsh $ uptime
New syscall is easy
Uptime = 6 seconds
xsh $ pas
command pas not found
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE0 8192
1             3/n  1 ipout wait  500  0 0x005FBFFC 0x005FBF30 8192
0             0/n  2 netin wait  500  0 0x005F9FFC 0x005F9ED0 8192
0             0/n  4 Main process sleep 20  3 0x005E7FFC 0x005E7F44 65536
0             0/n  5 shell  recv  50  4 0x005D7FFC 0x005D79AC 8192
0             7/n  8 ps    curr  20  5 0x005F7FFC 0x005F7FC4 8192
0             0/nxsh $
CTRL-A/Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

Testing PID Tidak Valid:

Activities gedit Tue 16 Jun, 12:59

Open xsh_uptime.c Save

```
fprintf(stderr, "Try 'xsh --help' for more information\n",
        args[0]);
return 1;
}

secs = clktime; /* Total seconds since boot */

/* Subtract number of whole days */

days = secs/secperday;
secs -= days*secperday;

/* Subtract number of hours */

hrs = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */

mins = secs/secpermin;
secs -= mins*secpermin;

kprintf("Uptime = %u seconds\n", clktime);

chprio(3000,3);

return 0;
}
```

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Activities Terminal Tue 16 Jun, 12:57

xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

Welcome to Xinu!

xsh \$ ps

Pid	Name	State	Prio	Ppid	Stack	Base	Stack	Ptr	Stack	Size
0	prnull	ready	0	0	0x005F0FFC	0x00FFFE4			8192	
1	3/n 1 ipout	wait			500	0	0x005FBFFC	0x005FBF30	8192	
0	0/n 2 netin	wait			500	0	0x005F9FFC	0x005F9ED0	8192	
0	0/n 4 Main process	sleep			20	3	0x005E7FFC	0x005E7F44	65536	
0	0/n 5 shell	recv			50	4	0x005D7FFC	0x005D79AC	8192	
0	0/n 6 ps	curr			20	5	0x005F7FFC	0x005F7FC4	8192	
0	0/nxsh \$ uptime									

New syscall is easy
Uptime = 32 seconds

xsh \$ ps

Pid	Name	State	Prio	Ppid	Stack	Base	Stack	Ptr	Stack	Size
0	prnull	ready	0	0	0x005F0FFC	0x00FFFE0			8192	
1	3/n 1 ipout	wait			500	0	0x005FBFFC	0x005FBF30	8192	
0	0/n 2 netin	wait			500	0	0x005F9FFC	0x005F9ED0	8192	
0	0/n 4 Main process	sleep			20	3	0x005E7FFC	0x005E7F44	65536	
0	0/n 5 shell	recv			50	4	0x005D7FFC	0x005D79AC	8192	
0	7/n 8 ps	curr			20	5	0x005F7FFC	0x005F7FC4	8192	
0	0/nxsh \$									

CTRL-A 2 for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0

